

Physics Midterm 2

1) $f = \frac{1}{1.33} \approx .75 \text{ Hz}$
 $.75 \cdot 60 = 45$
B

2)

3) **D**

4) $.12 \cdot (20.94)^2 \approx 52.5$
 $52.5 \times .001 = .53$
B

5) $v = 1.40 \cdot 30 = 42 \text{ m/s}$
C

Unit 5

1) **D**

2) a) $1000 \text{ N} \cdot .10 \text{ m} = 100 \text{ J}$
 b) Change in kinetic energy = 100 J
 c) $v = \sqrt{\frac{2 \times 100}{1.5}} = 36.5 \text{ m/s}$
 d) $\frac{36.5^2 \sin(90)}{9.8} = 136 \text{ m}$

3) a) $.05 \cdot 40 \cdot 9.8 \cdot 40 = 784 \text{ J}$
 b) $\frac{784}{30} = 26.1 \text{ W}$

4) $3w = 365 \cdot 24 h = 26,280 \text{ wh} = 26.28 \text{ kWh} \cdot .09 = 2.33$
D

Unit 6

1) C

2)

3) a) $4000 \cdot .2 = 800$

b) $\frac{800}{200} + 2.8 = 6.8 \text{ m/s}$

4) D

5) a) $\frac{30,000 \cdot .85}{149,000} = 0.182 \text{ m/s}$

b)

Unit 7

1) B

2)

Unit 8

1) a) $\frac{1}{2} \cdot .08^2 = .0032$

b) $.0032 \cdot 2 = .0064$

2) a) $.2 \cdot 12 = 2.4$